Lim Boon Hwee Matthew

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NS : ORD, March 2007

Professional Summary

Software engineer with 15+ years of experience building industrial automation, embedded systems, and energy management solutions — much of it self-taught on the job, from C# and PLC ladder diagrams to Node.js and OCPP-based EV charging systems. I thrive across the full development cycle: requirements, design, coding, testing, and deployment, having delivered end-to-end systems for ports, warehouses, and smart energy platforms across Singapore and the region. My career spans SME and R&D environments, giving me the range to work independently on greenfield projects as comfortably as within structured Agile teams. Currently deepening expertise in Industrial IoT and edge computing, and seeking roles that reward both technical depth and continued learning.

Highlights

• Experienced in **Energy Management, Smart Factory, and Industrial IoT** projects.
• Bridges **legacy PLC systems** with modern API-driven backends.
• Self-taught developer expanding from C# to Node.js, to embedded C++ and cloud systems.
• Interested in **industrial automation, and edge computing** roles.

Professional Experience

W2ESolutions Pte Ltd — Senior Software Developer

(Contractor supporting Panasonic R&D Center Singapore) Jan 2021 – Present
EV Energy Management System
• Developed C++ backend for EV Power balancing platform.
• Integration of EV chargers (EVSE), Battery Energy Storage System (BESS), and Solar PV
• Implemented OCPP 1.6J protocol for EVSE communications and Modbus TCP for BESS, Solar PV and Power Meter (PM) integration.
• Delivered to MVP3 Stage, with MVP2 deployed to a live facility.
• Established automated unit testing and integration test pipelines.
• Authored technical documentation for C++ components and applications.

Intelligent Lighting Control System
• Built Node.js backend for PLC-based intelligent solution.
• Integrated Modbus TCP communication to control PLC and DALI systems
• Integrated REST API and Websocket Secure interfaces for enable real-time monitoring and remote control of lighting systems.

Panasonic R&D Center Singapore — Senior Software Developer

Jun 2020 – Dec 2020

• Joined as direct hire before transitioning to contractor role. Contributed to the same project throughout both roles.
• Built C++ software components for Linux edge controllers, integrating SSL, JSON, IPC, and WebSocket Secure for industrial data collection.
• Established CI practices and testing foundations for the C++ codebase, supporting long-term maintainability

NEC Asia Pacific Pte Ltd — Software Developer (.NET)

Aug 2019 – Mar 2020

• Re-engineered legacy VB6 systems to C# for internal automation and government-sector prototypes.
• Migrated desktop workflows to WPF / WinForms architecture with configurable WebAPI integration.
• Delivered functional prototypes with SOLID principles, unit and integration testing, and technical documentation.

iDirect Asia Pte Ltd — Senior Engineer (Software)

Nov 2017 – Mar 2019

• Member of Continuous Engineering team maintaining iDirect's satellite-communication platform (C++/Linux).
• Contributed bug fixes and feature ports across multiple embedded system releases.
• Participated in Agile Scrum, performing code reviews, integration builds, and technical documentation.

NDI Automation Pte Ltd — Assistant Manager (Software Development)

Jun 2011 – Nov 2017

• Led design and deployment of industrial automation and monitoring software for major port and warehouse operators in Southeast Asia.
• Delivered end-to-end project lifecycle: requirements, system design, C#/PLC coding, testing, and commissioning.
• Created reusable C# frameworks for visualization, charting, and reporting to improve development efficiency.
• Integrated Siemens and Schneider PLCs with WinForms HMI systems using Modbus and socket communication.
• Provided on-site technical support, training, and documentation for deployed systems.

Selected Projects:

Crane Monitoring & Remote CMS Systems – Developed real-time monitoring and reporting platforms for container handling equipment at multiple regional ports.

ASRS (Automated Storage & Retrieval System) – Built warehouse-management software in C# with SQL and WinForms interfaces to PLC controls. Supported integration with RFID and image-capture modules for inventory verification.

Digital Hour Meter System – Designed PLC logic and C# interfaces for runtime tracking and diagnostics.

• Client: **Certis Cisco / Recall Singapore** – Warehouse Management System (2011 – 2012)
→ Migrated legacy WMS from C++ to C#, enhancing database reliability and network performance.
→ Provided ongoing maintenance and feature enhancements through 2015.

Earlier Roles

Teaching Assistant, National University of Singapore (2009 – 2011) – Labs and tutorials for Programming Methodology and Data Structures.

Education

Bachelor of Engineering (Computer Engineering) –
National University of Singapore
Second Upper Honours. CAP 4.04 / 5.00 (August 2007 – May 2011)

Technical Skills

Development Tools: Visual Studio, Visual Studio Code, Git, SVN, MobaXterm, Putty

Language and Frameworks (High Confidence): C, C++ (Linux), C# (Winforms, WPF, ASP.NET), Node.js

Languages and Frameworks (Working Knowledge): Python, JavaScript, HTML, XAML, PHP, Java, Visual Basic

Industrial Protocols and Communications Protocols: Modbus RTU, Modbus TCP, REST API, MQTT, WebSocket Secure (WSS), SSL/TLS, IPC, JSON

Operating Systems: Windows, Linux (Debian, Ubuntu, RedHat)

Methodologies & Design Patterns: Unit & Integration Testing, CI/CD